

KUUSAMO RUKATUNTURI, Finland

WMO: 028680

Lat: 66.1661N

Lon: 29.1517E

Elev: 363

StdP: 97.04

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.9	-24.9	0.4	-22.2	-22.3	0.5	-19.7	18.8	-9.0	17.3	-8.6	9.3	100	0.990

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	23.1	15.8	20.9	14.7	19.1	13.8	17.1	20.7	15.9	19.3	14.8	18.1	5.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	11.7	18.2	14.5	10.8	17.1	13.4	10.0	16.1	49.5	20.8	45.9	19.6	42.7	18.2	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	14.2	13.1	DB	-23.8	24.4	3.6	2.4	-26.4	26.1	-28.5	27.5	-30.5	28.8	-33.1	30.5
			WB	-23.4	18.2	3.6	1.7	-26.0	19.4	-28.1	20.4	-30.2	21.4	-32.8	22.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-11.4	-9.9	-6.9	-1.5	4.9	9.8	13.8	11.3	6.6	-0.5	-4.8	-7.8
	DBStd	9.38	4.90	5.20	3.92	3.72	5.10	4.58	4.05	3.71	3.39	3.43	3.69	4.41
	HDD10.0	3792	663	557	524	344	173	59	12	28	110	326	443	552
	HDD18.3	6572	922	790	783	594	416	259	149	219	353	583	693	810
	CDD10.0	274	0	0	0	0	16	52	129	69	7	0	0	0
	CDD18.3	13	0	0	0	0	0	3	7	3	0	0	0	0
	CDH23.3	38	0	0	0	0	1	11	18	8	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	8.5	8.4	8.2	7.5	7.2	6.5	6.0	6.3	6.9	7.4	8.2	8.4
Precipitation	PrecAvg	569	38	31	30	29	46	64	82	63	52	49	44	41
	PrecMax	731	73	77	52	44	80	127	142	139	97	89	99	73
	PrecMin	444	17	6	9	9	17	19	30	25	9	8	8	10
	PrecStd	78	15	17	12	11	16	29	26	26	24	22	20	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	0.2	2.9	11.2	21.2	24.8	25.3	24.6	16.4	8.6	4.1	-0.1
		MCWB	-0.5	0.2	0.8	6.0	11.6	15.9	17.1	17.9	12.4	8.4	3.1	-0.1
	2%	DB	-2.1	-0.7	0.6	7.6	18.9	22.2	23.5	20.7	14.1	6.2	2.2	-0.7
		MCWB	-2.2	-0.7	-0.6	3.6	11.4	14.7	16.3	15.4	11.6	5.9	1.9	-0.7
	5%	DB	-3.3	-1.7	-0.6	4.9	15.9	18.9	21.7	18.6	12.4	5.2	0.9	-1.4
		MCWB	-3.4	-1.8	-1.3	2.0	10.0	13.0	15.1	14.4	10.3	4.9	0.8	-1.4
	10%	DB	-4.7	-2.9	-1.6	3.4	12.7	16.7	19.9	16.6	11.1	4.1	-0.2	-2.2
		MCWB	-4.9	-3.0	-2.1	1.1	7.9	11.4	14.6	13.3	9.6	3.8	-0.2	-2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	0.2	1.2	6.2	13.1	17.4	18.7	18.5	13.5	8.4	3.4	-0.1
		MCDB	-0.5	0.2	2.4	9.5	17.6	21.4	22.2	23.2	15.2	8.6	3.4	-0.1
	2%	WB	-2.1	-0.7	-0.3	4.3	11.7	15.6	17.5	16.2	11.8	6.1	1.8	-0.8
		MCDB	-2.1	-0.7	0.2	7.1	17.7	20.4	20.9	19.1	13.2	6.1	1.9	-0.8
	5%	WB	-3.4	-1.7	-1.1	2.5	10.3	13.9	16.5	14.9	10.8	4.9	0.5	-1.5
		MCDB	-3.2	-1.7	-0.7	4.3	15.6	18.3	19.9	17.5	11.9	5.1	0.6	-1.5
	10%	WB	-4.9	-3.0	-2.0	1.5	8.4	12.2	15.5	13.9	9.9	3.8	-0.4	-2.3
		MCDB	-4.8	-2.9	-1.7	2.9	12.4	15.3	18.9	16.1	10.8	4.0	-0.3	-2.3
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.4	4.8	4.8	6.4	6.4	6.4	5.5	4.3	3.1	3.4	3.6
		MCDBR	4.9	4.3	4.8	6.5	10.1	8.8	8.0	7.6	5.8	4.3	4.0	3.4
		MCWBR	5.1	4.3	4.4	4.0	5.6	4.5	4.1	4.1	3.9	3.9	3.8	3.6
	5% WB	MCDBR	4.8	4.3	4.5	5.7	9.8	8.4	7.4	6.6	5.3	4.2	3.5	3.6
		MCWBR	5.1	4.3	4.2	3.8	5.7	5.0	4.5	4.3	3.9	4.0	3.5	3.7
Clear-Sky Solar Irradiance	taub		0.189	0.256	0.289	0.321	0.334	0.334	0.350	0.330	0.307	0.278	0.350	0.190
	taud		1.923	2.112	2.170	2.250	2.460	2.496	2.467	2.498	2.546	2.366	2.731	1.947
	Ebn at Noon		241	599	767	839	870	878	848	832	774	592	256	30
	Edh at Noon		17	54	85	102	93	92	92	80	61	43	18	4
All-Sky Solar Radiation	RadAvg		0.09	0.59	2.05	3.83	4.74	5.06	4.71	3.43	1.84	0.61	0.13	0.01
	RadStd		0.01	0.11	0.19	0.37	0.44	0.37	0.48	0.34	0.19	0.10	0.02	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page